

Proposal: Smart-City Traffic Analysis using NYC Taxi Trips

CS F415 Data Mining — Group 5

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1. Executive Summary A Municipal Corporation requires data-driven interventions to reduce traffic congestion and improve transit planning. We analyzed the NYC Taxi Trip Duration dataset with **1,458,644** trips from **January 1, 2016 to June 30, 2016**. The dataset contains 11 features including pickup/dropoff timestamps, GPS coordinates, passenger count, and trip duration. This proposal defines three linked data mining tasks: (1) *trip duration prediction* for travel-time estimation and anomaly detection, (2) *spatial demand clustering* for high-demand zone discovery, and (3) *congestion-level classification* for dynamic transit scheduling and traffic response.

2. Data Exploration Summary The dataset is complete (no missing values) and temporally consistent (no pickup-after-dropoff records; no mismatch between computed and provided duration). Basic profiling indicates heavy right-skew in trip duration and strong time-of-day effects in both trip counts and speed.

Key Statistics

Metric	Value
Total records	1,458,644
Features	11
Date range	2016-01-01 to 2016-06-30
Mean / median trip duration	959.49 s / 662 s
Trip duration standard deviation	5,237.43 s
Mean / median Haversine distance	3.441 km / 2.094 km
Median estimated speed (overall)	12.792 km/h
Median speed: peak vs off-peak	11.851 vs 19.180 km/h
Trips during peak hours (7–9, 17–19)	30.69%
Vendor split (1 / 2)	46.50% / 53.50%

Data Quality Findings

- Coordinates outside NYC bounds (40.4–41.0 N, -74.3 to -73.7 W): **985 trips** (0.0675%); invalid pickups: 247, invalid dropoffs: 971.
- Short-duration anomalies (<60 s): **8,595 trips** (0.5892%); long-duration anomalies (>24 h): **4 trips**.
- Passenger-count anomalies: **60** zero-passenger trips; **5** trips with >6 passengers.
- Estimated speed anomalies (>100 km/h): **175 trips**; zero-distance trips: **5,897**.

Notable Patterns

- Demand shows clear commute peaks in morning and evening hours.
- Median speed drops during peak windows, indicating higher congestion.
- Spatial demand is concentrated around core Manhattan and major airport corridors.

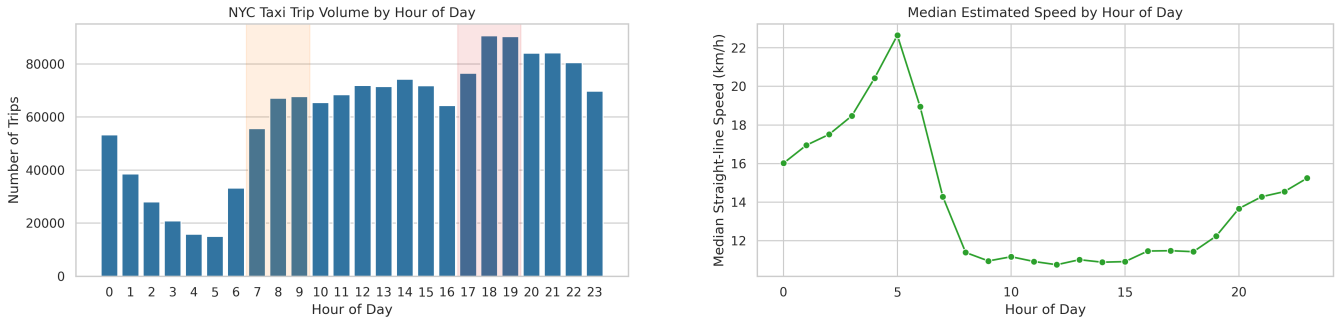


Figure 1: Left: hourly trip demand reveals AM/PM peak periods. Right: median straight-line speed dips during rush hours, a direct congestion proxy.

3. Problem Formulation

Task 1: Trip Duration Prediction (Supervised Regression)

Input $\rightarrow$ *Output*: Temporal features (hour, weekday, month), pickup/dropoff coordinates, vendor/passenger fields, Haversine distance $\rightarrow$ predicted trip duration (seconds).

- Travel-time forecasting for route planning and passenger information systems
- Real-time incident flagging when actual duration greatly exceeds prediction
- Dynamic bus/subway capacity allocation on parallel corridors
- Evidence base for time-zone congestion pricing

Planned models: Linear Regression, Random Forest, Gradient Boosting.

Task 2: High-Demand Zone Identification (Unsupervised Clustering)

Input $\rightarrow$ *Output*: Pickup coordinates (optionally enriched with demand counts and mean duration) $\rightarrow$ cluster labels representing geographic demand zones.

- Identify underserved high-demand areas for transit augmentation
- Prioritize zones for traffic signal and lane interventions
- Guide park-and-ride or bike-share station placement
- Inform zone-based congestion pricing and infrastructure spending

Planned models: K-Means **vs.** DBSCAN (algorithm comparison).

Task 3: Congestion Level Classification (Supervised Multi-class)

Input $\rightarrow$ *Output*: Zone ID, hour, weekday, historical trip count, and average speed $\rightarrow$ congestion class {High, Medium, Low}. Labels derived via speed-percentile thresholds across zone-time buckets.

- Dynamic public-transit service frequency control
- Commuter-facing congestion alerts and route suggestions
- Optimized staffing and operations planning for transit agencies
- Data-backed policy support for staggered work hours

Planned models: Logistic Regression, Random Forest Classifier.

4. Expected Outcomes & Next Steps Phase 2 (Weeks 9–15) will operationalize these tasks through preprocessing, feature pipeline finalization, weather-data integration, model comparison, and failure analysis. Expected outcomes are actionable recommendations such as targeted peak-hour bus deployment, high-friction zone interventions, and evidence-based congestion pricing policies linked to measured speed and demand patterns.